

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 3

1. Some examples for the discrete random variable:

1. **Bernoulli Distribution**

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}$$

2. **Binomial Distribution**

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, \dots, n$$

3. **Geometric Distribution**

$$P(X = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots$$

4. **Poisson Distribution**

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

2. **(Review of continuous random variable)**

- Definition of CDF $F(x) = \mathbb{P}(X \leq x)$ and PDF $f(x)$. The PDF is defined as the function satisfying

$$\forall -\infty \leq a \leq b \leq \infty, \quad \mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx.$$

- Their properties:

- CDF: continuous, nondecreasing, $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.
- PDF: $\int f(x) dx = 1$, $f(x) \geq 0$.

- The relationship between CDF and PDF: $F(x) = \int_{-\infty}^x f(t) dt$; $f(x) = F'(x)$.

- Expectation and Variance:

- $\mathbb{E}[g(X)] = \int g(x) f(x) dx$.
- $\text{var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

- Some examples for the continuous random variable: Uniform $f(x) = \frac{1}{b-a} 1\{a \leq x \leq b\}$; Exponential $f(x) = \lambda e^{-\lambda x} 1\{x \geq 0\}$; Standard Gaussian $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

3. **(An Illustration: Linear Transformation of A Random Variable)**

- X continuous random variable, CDF $F(x)$, PDF $f(x)$.
- $Y = aX + b$ for some constant a, b . ($a \neq 0$) $\mathbb{E}[Y] = a\mathbb{E}[X] + b$; $\text{var}(Y) = a^2 \text{var}(X)$.
- CDF, PDF of Y ?

– **Solution:**

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(aX + b \leq y) = \mathbb{P}(X \leq \frac{y-b}{a}) = F_X(\frac{y-b}{a})$$
$$f_Y(y) = F'_Y(y) = \frac{1}{a} f_X(\frac{y-b}{a})$$

- Example of Gaussian random variable $X \sim \mathcal{N}(\mu, \sigma^2)$.

- $Z \sim N(0, 1)$ standard Gaussian, $X = \sigma Z + \mu \sim \mathcal{N}(\mu, \sigma^2)$.
- $X \sim \mathcal{N}(\mu, \sigma^2)$, $\frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$

4. **(Continuous Random Variable)** The lifetime of a system is modeled by a random variable X with PDF

$$f(x) = \begin{cases} kxe^{-x/2} & x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Question:

- (a) Find the corresponding CDF of random variable X .
- (b) Compute the value $\mathbb{P}(X > 5)$.
- (c) Compute the expected lifetime $\mathbb{E}[X]$.

• **Solution:**

- (a) Integrate the PDF, we have

$$\int_0^\infty f(x)dx = 1,$$

so $k = 1/4$. By definition of the CDF, for every $c > 0$,

$$F(c) = \mathbb{P}(X \leq c) = \int_0^c f(x)dx = 1 - \left(1 + \frac{x}{2}\right)e^{-x/2},$$

and for every $c \leq 0$, $F(c) = 0$.

- (b)

$$\mathbb{P}(X > 5) = 1 - F(5) = \left(1 + \frac{5}{2}\right)e^{-5/2}.$$

- (c)

$$\mathbb{E}[X] = \int_0^\infty \frac{1}{4}x^2e^{-x/2}dx = \frac{1}{2}\mathbb{E}[Y^2] = \frac{1}{2}(2^2 + 2^2) = 4,$$

where Y is a random variable with exponential distribution $Y \sim \exp(\frac{1}{2})$.